

- 5 *Listeria monocytogenes* is a bacterium that contaminates food and causes the disease listeriosis.

The photograph shows a colony of these bacteria, as seen using an electron microscope.



(Source: © BSIP SA / Alamy Stock Photo)

- (a) The bacterium labelled X on the photograph is $0.5\ \mu\text{m}$ in length.

(i) Calculate the magnification of this image.

(1)

Answer

- (ii) State why a light microscope cannot be used to view these bacteria.

(1)

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- (b) The photograph shows a bottle of oil extracted from the plant oregano. This plant is native to Mediterranean and western Asian countries.



(Source: © Frank Hecker / Alamy Stock Photo)

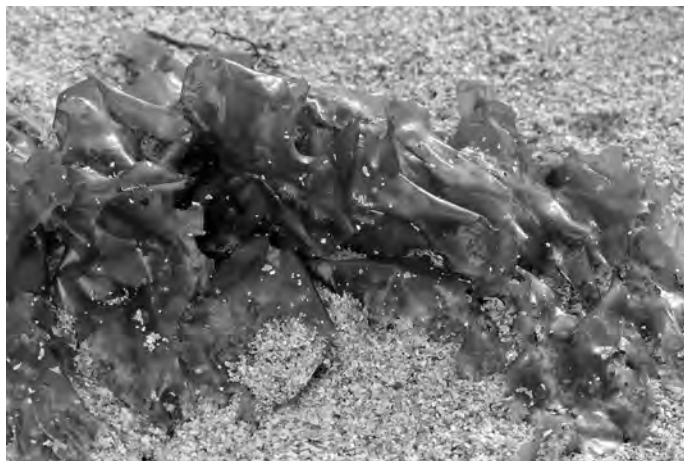
Oregano oil contains antimicrobial substances.

This oil can be used in the production of plastic used to package food.

Explain why this packaging may be chosen over traditional oil-based plastic packaging.

(2)

(c) The photograph shows green algae called sea lettuce (*Ulva lactuca*).



(Source: © Arterra Picture Library / Alamy Stock Photo)

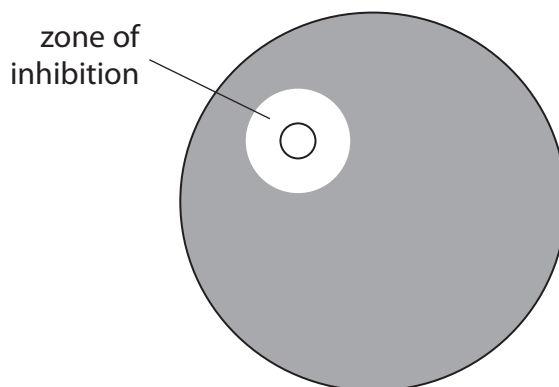
Sea lettuce contains chemicals called fucoxanthins that have antimicrobial properties.

The antimicrobial properties of these chemicals were investigated.

Discs containing fucoxanthin chemical were placed onto four agar plates each seeded with a different type of bacteria.

These plates were then incubated for 24 hours at 25 °C and the zone of inhibition measured.

The diagram shows the zone of inhibition.



The table shows the results of this investigation.

Type of bacteria	Mean diameter of the zone of inhibition / mm	Range / mm
<i>E. coli</i>	10.2	± 0.72
<i>S. aureus</i>	11.1	± 0.63
<i>L. monocytogenes</i>	6.0	± 0.41
<i>P. aeruginosa</i>	7.5	± 0.55



(i) Which type of bacteria was least affected by fucoxanthin?

(1)

- ☐ **A** *E. coli*
- ☐ **B** *S. aureus*
- ☐ **C** *L. monocytogenes*
- ☐ **D** *P. aeruginosa*

(ii) The table shows the mean and the range of the diameter of the zone of inhibition for each type of bacteria.

Calculate the maximum difference in the diameter of the largest and smallest zones of inhibition.

(1)

Answer mm

(d) Drugs containing these chemicals must be tested for safety before they can be approved and used to treat humans.

Three-phase testing can be used to check for the safety and effectiveness of these chemicals.

Describe the roles of phase I and phase II.

(3)

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(Total for Question 5 = 9 marks)